



Tutorial 9C: Differential Equations

Section A (Basic Questions)

1 Solve the following differential equations.

- (a) $\frac{dy}{dx} = \sec^2 x$,
- (b) $\frac{dy}{dx} = y^2 + a^2$, where a is a positive constant,
- (c) $\frac{dy}{dx} = xe^{-y}$.

Solution:	
(a)	$\frac{dy}{dx} = \sec^2 x$ $\int 1 \, dy = \int \sec^2 x \, dx$ $y = \tan x + c, \text{ where } c \in \mathbb{R}$
(b)	$\frac{dy}{dx} = y^2 + a^2$ $\int \frac{1}{y^2 + a^2} \, dy = \int 1 \, dx$ $\frac{1}{a} \tan^{-1} \frac{y}{a} = x + c, \text{ where } c \in \mathbb{R}.$ $\tan^{-1} \frac{y}{a} = ax + ac$ $\frac{y}{a} = \tan(ax + ac)$ $y = a \tan(ax + A), \text{ where } A = ac \in \mathbb{R}.$ You can leave your answer in this form if the question does not require you to make y the subject.
(c)	$\frac{dy}{dx} = xe^{-y}$ $\int e^y \, dy = \int x \, dx$ $e^y = \frac{x^2}{2} + c$ $y = \ln\left(\frac{x^2}{2} + c\right), \text{ where } c \in \mathbb{R}$

2 Solve the following differential equations using the given substitutions.

(a) $\frac{dy}{dx} + (1 + (x - y)^2) \cos x = 1$, by substituting $v = x - y$,

(b) $\frac{dy}{dx} = \sec^2(y - x - 3)$, by substituting $u = y - x - 3$, given that $y = \frac{\pi}{4}$ when $x = -3$.

Solution:

(a)

$$v = x - y \Rightarrow \frac{dv}{dx} = 1 - \frac{dy}{dx}$$

Substitute into the differential equation $\frac{dy}{dx} + (1 + (x - y)^2) \cos x = 1$,

$$1 - \frac{dv}{dx} + (1 + v^2) \cos x = 1$$

$$\frac{dv}{dx} = (1 + v^2) \cos x$$

$$\int \frac{1}{1 + v^2} dv = \int \cos x \, dx$$

$$\tan^{-1} v = \sin x + c, \text{ where } c \in \mathbb{R}$$

$$\tan^{-1}(x - y) = \sin x + c$$

$$\therefore y = x - \tan(\sin x + c).$$

(b)

$$u = y - x - 3 \Rightarrow \frac{du}{dx} = \frac{dy}{dx} - 1$$

Substitute into the differential equation $\frac{dy}{dx} = \sec^2(y - x - 3)$,

$$\frac{du}{dx} + 1 = \sec^2 u$$

$$\frac{du}{dx} = \sec^2 u - 1$$

$$\int \frac{1}{\sec^2 u - 1} du = \int 1 \, dx$$

$$\int \cot^2 u \, du = \int 1 \, dx$$

$$\int (\operatorname{cosec}^2 u - 1) du = \int 1 \, dx$$

$$-\cot u - u = x + c, \text{ where } c \in \mathbb{R}$$

$$-\cot(y - x - 3) - (y - x - 3) = x + c$$

$$\cot(y - x - 3) = -y + d, \text{ where } d \in \mathbb{R}$$

When $x = -3$, $y = \frac{\pi}{4}$ and thus $d = \cot\left(\frac{\pi}{4}\right) + \frac{\pi}{4} = 1 + \frac{\pi}{4}$.

$$\therefore \cot(y - x - 3) = -y + 1 + \frac{\pi}{4}.$$

- 3 The curve C , which does not pass through the origin, is such that at any point on the curve the tangent is perpendicular to the line drawn from the origin to the point. Show that the differential equation $y \frac{dy}{dx} + x = 0$ is satisfied by all points (x, y) on C . Given that C passes through the point with coordinates $(3, 4)$, find the equation of C .

Solution:

At any point (x, y) on C , gradient of the tangent at this point $= \frac{dy}{dx}$.

Gradient of the line from the origin to the point $(x, y) = \frac{y}{x}$.

Since the tangent is perpendicular to the line,

$$\frac{dy}{dx} \left(\frac{y}{x} \right) = -1$$

$$y \frac{dy}{dx} = -x$$

$$y \frac{dy}{dx} + x = 0 \quad (\text{shown})$$

$$y \frac{dy}{dx} = -x$$

$$\int y \, dy = \int -x \, dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + k$$

$$y^2 = -x^2 + k', \quad \text{where } k' = 2k \text{ is a real constant}$$

Given that C passes through $(3, 4)$, $k' = 25$.

Hence equation of C is $y^2 + x^2 = 25$.

- 4 A bottle containing liquid is taken from a refrigerator and placed in a room where the temperature is a constant 20°C . As the liquid warms up, the rate of increase of its temperature $\theta^{\circ}\text{C}$ after time t minutes is proportional to the temperature difference $(20 - \theta)^{\circ}\text{C}$. Initially the temperature of the liquid is 10°C and the rate of increase of the temperature is 1°C per minute. By setting up and solving a differential equation, show that $\theta = 20 - 10e^{-\frac{1}{10}t}$. [7]

Find the time it takes the liquid to reach a temperature of 15°C , and state what happens to θ for large values of t . Sketch a graph of θ against t . [4]

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Solution:

$$\frac{d\theta}{dt} = k(20 - \theta)$$

Given that when $\theta = 10$, $\frac{d\theta}{dt} = 1$

$$1 = k(20 - 10)$$

$$k = \frac{1}{10}$$

$$\therefore \frac{d\theta}{dt} = \frac{1}{10}(20 - \theta)$$

$$\int \frac{1}{20 - \theta} d\theta = \frac{1}{10} \int 1 dt$$

$$-\ln|20 - \theta| = \frac{1}{10}t + C', \text{ where } C' \in \mathbb{R}$$

$$\ln|20 - \theta| = -\frac{1}{10}t + C, \text{ where } C = -C'$$

$$|20 - \theta| = e^{-\frac{1}{10}t + C}$$

$$20 - \theta = \pm e^{-\frac{1}{10}t} e^C = A e^{-\frac{1}{10}t} \text{ where } A = \pm e^C$$

$$\theta = 20 - A e^{-\frac{1}{10}t}$$

Given that when $t = 0$, $\theta = 10$,

$$10 = 20 - A \Rightarrow A = 10$$

$$\therefore \theta = 20 - 10e^{-\frac{1}{10}t} \text{ (shown)}$$

Alternatively,

$$\ln|20 - \theta| = -\frac{1}{10}t + C$$

$$\ln(20 - \theta) = -\frac{1}{10}t + C,$$

(since $\theta = 10$ when $t = 0 \Rightarrow |20 - \theta| = 20 - \theta$)

$$20 - \theta = B e^{-\frac{1}{10}t}, B = e^C \in \mathbb{R}^+.$$

$$\theta = 20 - B e^{-\frac{1}{10}t}$$

This is the general solution. Read the question carefully and sift out initial conditions and use them to find the particular solution.

$$\theta = 20 - 10e^{-\frac{1}{10}t}$$

At $\theta = 15$,

$$15 = 20 - 10e^{-\frac{1}{10}t}$$

$$10e^{-\frac{1}{10}t} = 5$$

$$-\frac{1}{10}t = \ln \frac{1}{2}$$

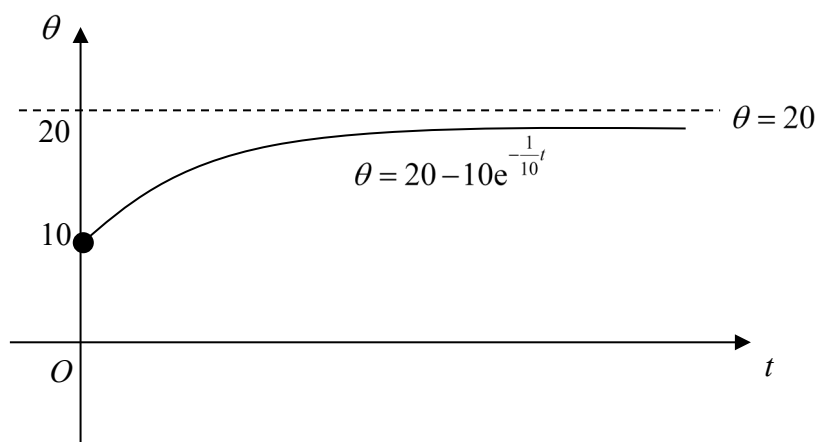
$$t = -10 \ln \frac{1}{2} = 10 \ln 2 = 6.93 \text{ min}$$

Cambridge report commented that “it was unusual to see time left as $10 \ln 2$ ”, which suggested that the answer should be evaluated rather than left in exact form for contextual question.

As $t \rightarrow \infty$, $e^{-\frac{1}{10}t} \rightarrow 0$. $\therefore \theta \rightarrow 20$.

i.e. the temperature of the liquid **increases** and **approaches** room temperature, i.e. 20°C , for large values of t .

Pay attention to what is expected in answering such questions (key words in red)



Remark: Note that we only sketch for $t \geq 0$.

THE END